DIGITAL DICE USING DECADE COUNTER & 7 SEGMENT DISPLAY

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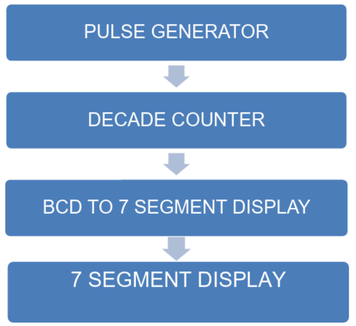
OBJECTIVES

- ❑ To design and implement a digital dice circuit.
- ❑ To understand the practical application of counters and display devices.
- ❑ To display numbers from 1 to 6 accurately using digital components.

METHODOLOGY

- ❑ Designed circuit using IC 4017 decade counter
- ❑ Generated clock pulses using pulse generator
- ❑ Used BCD-to-7 segment decoder for output conversion
- ❑ Displayed output on seven segment display

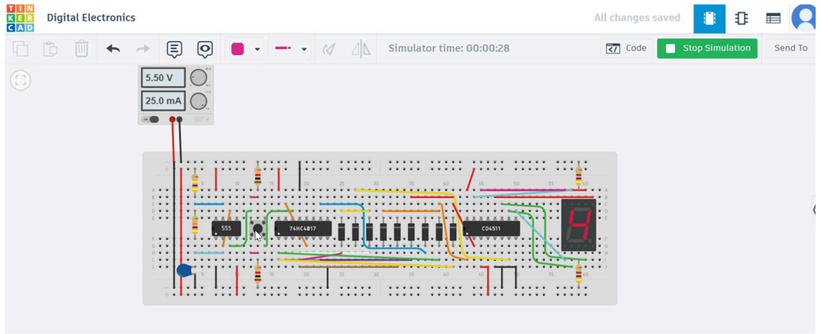
BLOCK DIAGRAM



RELEVANCE

- ❑ Demonstrates practical use of digital electronics concepts
- ❑ Replaces mechanical dice with an electronic system
- ❑ Useful for gaming devices and random number generation
- ❑ Helps students understand counters and display interfacing

SIMULATION DIAGRAM



CONCLUSION

- ❑ The digital dice provides a reliable electronic alternative to mechanical dice.
- ❑ The project demonstrates effective use of counters and display units.
- ❑ It is simple, cost-effective, and suitable for learning digital electronics.